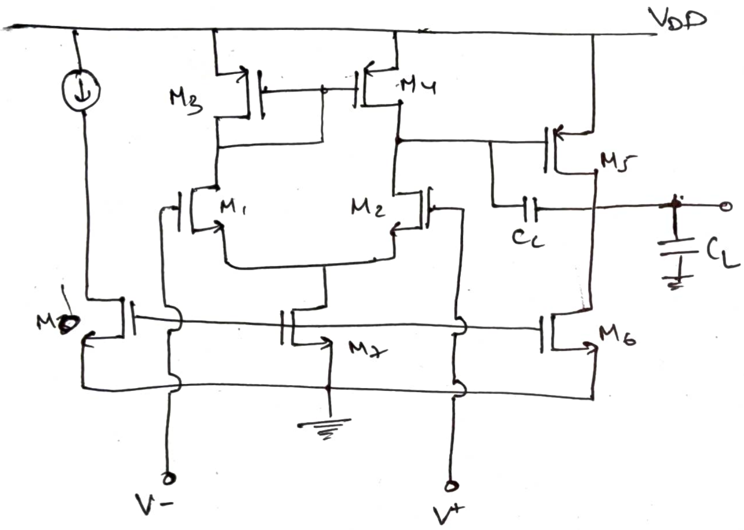
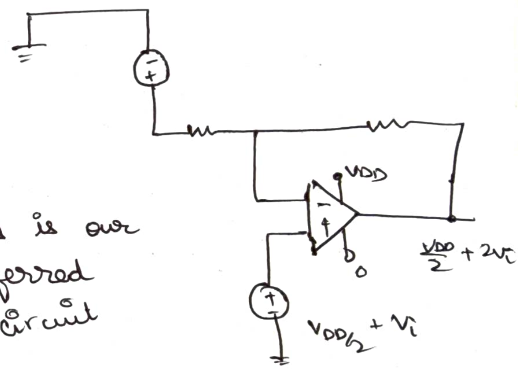


Circuit Diagram

Assumptions & Given Informations

This is our preferred circuit



Specifications

i) atleast 40 dB (let's take 60 dB) of DC gain

ii) GBW = 5 MHz

iii) PM $\geq$ 60°

iv) Slew Rate $\geq$ 10V/μsec

v) ICMR (+1) = 1.6V

ICMR (-1) = 0.8V

vi) Let's take C_L = 10pF

vii) V_{DD} = 1.8V

viii) $\mu_n C_{ox}$ = 200 μA/V² (initially)

$\mu_p C_{ox}$ = 100 μA/V² (initially)

Calculations :-

Slew Rate $\frac{I_5}{C_c}$

and for $\geq$ 60° Phase Margin

$C_c = 0.4 C_L = 4pF$

$I_7 = 10 \times 10^6 \times 4 \times 10^{-12}$

$\boxed{I_7 = 40 \mu A}$

M_1 & M_2

$g_{m_1} = GBW \cdot C_c \cdot 2\pi$

$g_{m_1} = 5 \times 10^6 \times 2\pi \times 4 \times 10^{-12}$

$g_{m_1} = 40\pi \mu S$

$g_{m_1} \approx 125 \mu S$

$g_{m_1} = \sqrt{I_{D_7} \mu_n C_{ox} \left(\frac{W}{L}\right)}$

$$\frac{W}{L} = \frac{(g_{m1})^2}{\mu_n C_{ox} I_{D7}} = \frac{(125)^2}{230 \times 40} = 1.69$$

$$\boxed{\frac{W}{L} = 1.69}$$

From the Rough slide we get

$$\sqrt{\frac{I_{D5}}{B_3}} = V_{DD} - |V_{th3}| + V_{th1} - V_{in})_{max}$$

$$\left(\frac{W}{L}\right)_{3,4} = \frac{I_s}{\mu_p C_{ox} \left(V_{DD} - (V_{th3}) + V_{th1} - V_{in\,max}\right)^2}$$

$$\left(\frac{W}{L}\right)_{3,4} = \frac{40\,\mu A}{100\,\frac{\mu A}{V}\,(1.8 - 0.37 + 0.31 - 1.6)^2}$$

$$\boxed{\left(\frac{W}{L}\right)_{3,4} \Rightarrow 20.40}$$

Negative CMRR use for the second one.

$$\underbrace{V_{dsat7}}_{\text{Overdrive voltage}} = 0.8 - 0 - \sqrt{\frac{I_{D7}}{B_1}} - V_{th}$$

$$V_{dsat7} = 0.8 - \sqrt{\frac{40}{230 \times 1.69}} - 0.37$$

$$V_{dsat7} = 0.8 - 0.37 - 0.1$$

$$\boxed{V_{dsat7} = 0.12\,V}$$

$$\left(\frac{W}{L}\right)_7 = \frac{2 I_{D7}}{\mu_n C_{ox} V_{Dsat}}$$

$$\left(\frac{W}{L}\right)_7 = \frac{2 \times 40\mu A}{230\frac{\mu A}{V} \times 0.12V}$$

$$\boxed{\left(\frac{W}{L}\right)_7 = \left(\frac{W}{L}\right)_6 = 2.9}$$

M_0 and M_5 from current mirror but given we have to have $1/10^{th}$ of the I_7 in the $\left(\frac{W}{L}\right)_{M_0}$

$$\left(\frac{W}{L}\right)_{M_0} = \frac{1}{10}(2.9) = \underline{\underline{0.29}}$$

$$\frac{12\mu}{L} = 0.29$$

$$\boxed{L = \frac{12}{0.29}\mu}$$ Rough

for Reasonable Phase Margin ($\sim 60°$)

$$g_{m5} = 2.2\, g_{m4}\left(\frac{C_L}{C_C}\right)$$

and $g_{m5} >> 10 g_{m1}$

$$V_{SG4} = V_{SG} = 5$$

$$\left(\frac{W}{L}\right)_5 = \left(\frac{W}{L}\right)_4 \left(\frac{g_{m5}}{g_{m4}}\right)$$

calculations of I_{D6}

$$g_{m6} = \sqrt{2\mu_p C_{ox}\left(\frac{W}{L}\right)_6 I_{D6}}$$

$$g_{m6}^2 = 2\mu_p C_{ox}\left(\frac{W}{L}\right)_6 I_{D6}$$

$$I_{D6} = \frac{(g_{m6})^2}{2\mu_p C_{ox}\left(\frac{W}{L}\right)_6}$$

$\left(P_2 = \frac{g_{m5}}{2\pi C_L}\right)$ Comes purely from the second stage

$GBW = 5\,MHz$

$P_2 \geq 2.2 \times 5\,MHz = 11\,MHz$

$P_2 \approx \frac{g_{m5}}{2\pi C_L}$

$11 \times 10^6 = \frac{g_{m5}}{2\pi \times (10 \times 10^{-6})}$

$g_{m5} \geq 691\,/\mu s$

Now since overdrive in both are same:-

$(V_{GS} - V_{TH})$ is same for both M_4 & M_5;

$g_{m4} = \sqrt{2\mu_p C_{ox} (W/L)_4 I_{D4}}$

$g_{m4} = \sqrt{2 \times 100/\mu \times 20.45\ (20)\mu}$

$\boxed{g_{m4} = 286\,/\mu s}$

$\Delta V_{D4} = \frac{2 I_{D4}}{g_{m4}} = \frac{2(20)/\mu A}{343\,/\mu s}$

$\boxed{\Delta V_{D4} = 0.14V}$

So for better phase margin we already got $g_{m5} \geq 691\,/\mu s$

we must have $\Delta V_5 = \Delta V_3 = 0.14V$

$$g_{m5} = \frac{2I_{D5}}{V_{ov5}}$$

$$I_{D5} = \frac{691 \mu \times 0.14}{2}$$

$$\boxed{I_{D5} = 48.37 \mu A}$$

Lets Assume ($\simeq 55 \mu A$)

$$\left(\frac{W}{L}\right)_5 = 20.45 \times 2.75 = 56.2375$$

$$\boxed{\begin{array}{l} W = 25.551 \mu m \\ L = 0.45 \mu m \end{array}}$$

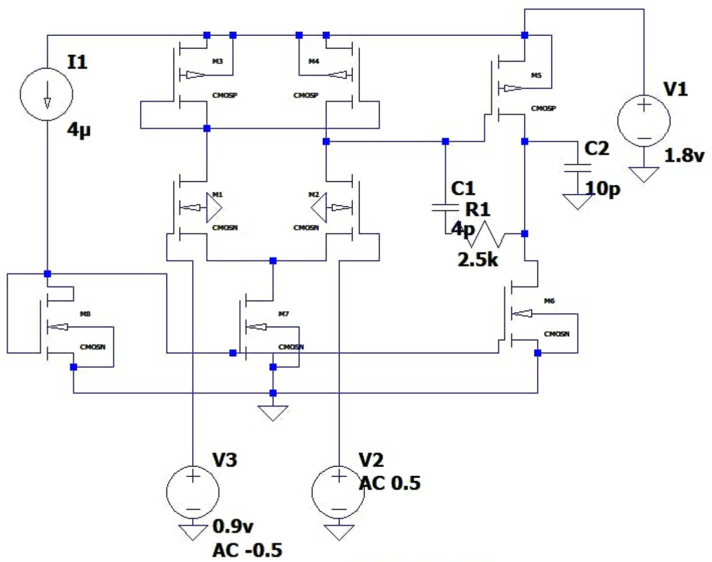
Now

M6 must be able to pull this much current

$\left(\frac{55}{4}\right) \rightarrow$ multiplier

$$\frac{55}{4} \times 1.69 \approx \frac{55}{4} \times 2 = \underline{\underline{27.5}}$$

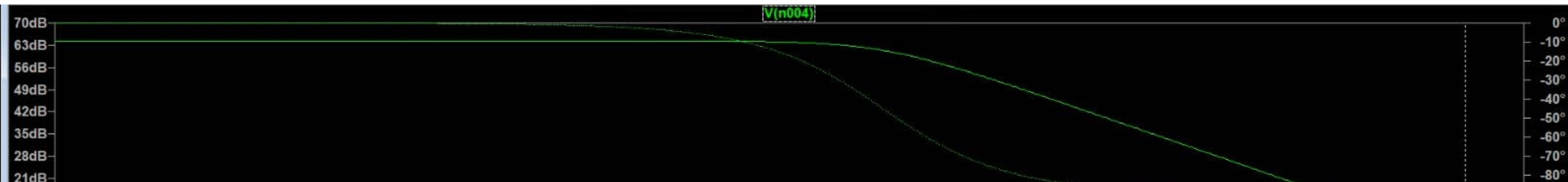
.include tsmc018.lib



.dc V2 0.89 0.91 10U

898.71233mV

.ac dec 100 0.1 10Meg



V(n004)
70dB
63dB
56dB
49dB
42dB
35dB
28dB
21dB
14dB
7dB
0dB
-7dB
-14dB
0°
-10°
-20°
-30°
-40°
-50°
-60°
-70°
-80°
-90°
-100°
-110°
-120°
-130°
-140°
100mHz
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz

Cursor 1

V(n004)

Freq:	4.7962176MHz	Mag:	-252.70729mdB
		Phase:	-116.34628°
		Group Delay:	12.406939ns

Cursor 2

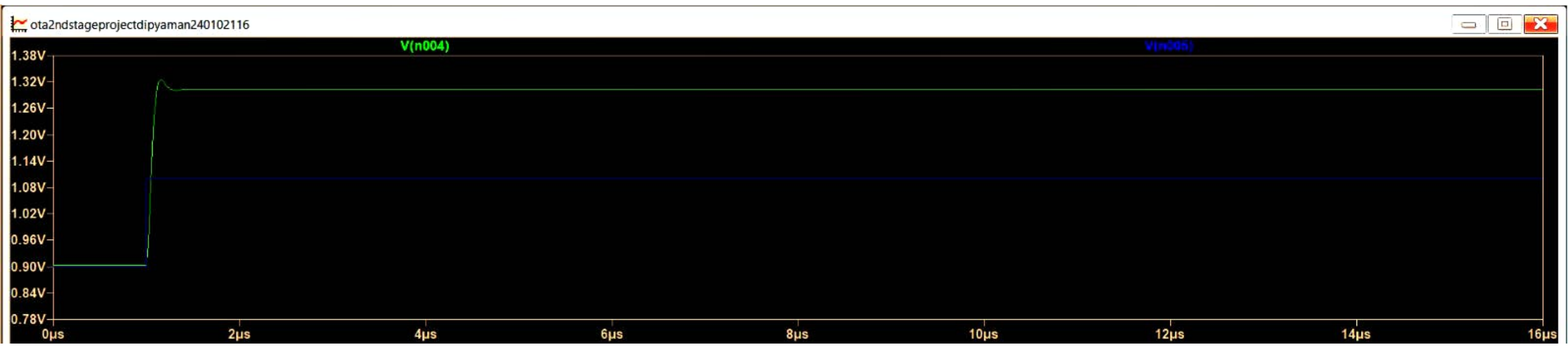
Freq:	-- N/A--	Mag:	-- N/A--
		Phase:	-- N/A--
		Group Delay:	-- N/A--

Ratio (Cursor2 / Cursor1)

Freq:	-- N/A--	Mag:	-- N/A--
		Phase:	-- N/A--
		Group Delay:	-- N/A--

V(n001):	1.8	voltage
V(n002):	1.15744	voltage
V(n003):	1.19874	voltage
V(n004):	0.902604	voltage
V(n005):	0.9	voltage
V(n006):	0.902604	voltage
V(n007):	0.901302	voltage
V(n008):	0.0849597	voltage
V(n009):	0.557781	voltage
V(n010):	0.9	voltage
I(C1):	1.18453e-24	device_current
I(C2):	9.02604e-24	device_current
I(I1):	4e-06	device_current
I(R1):	0	device_current
I(R2):	1.30188e-10	device_current
I(R3):	1.30188e-10	device_current
I(V1):	-0.000106646	device_current
I(V2):	0	device_current
I(V4):	1.30188e-10	device_current
Ib(M1):	-1.26203e-12	device_current
Ib(M2):	-1.30332e-12	device_current
Ib(M3):	6.52559e-13	device_current
Ib(M4):	6.11263e-13	device_current
Ib(M5):	9.07396e-13	device_current
Ib(M6):	-9.12604e-13	device_current
Ib(M7):	-9.45853e-14	device_current
Ib(M8):	-5.67781e-13	device_current

Ib(M8):	-5.67781e-13	device_current
Id(M1):	2.79511e-05	device_current
Id(M2):	2.78263e-05	device_current
Id(M3):	2.79511e-05	device_current
Id(M4):	2.78263e-05	device_current
Id(M5):	4.68687e-05	device_current
Id(M6):	4.68686e-05	device_current
Id(M7):	5.57773e-05	device_current
Id(M8):	4e-06	device_current
Ig(M1):	0	device_current
Ig(M2):	0	device_current
Ig(M3):	-0	device_current
Ig(M4):	-0	device_current
Ig(M5):	-0	device_current
Ig(M6):	0	device_current
Ig(M7):	0	device_current
Ig(M8):	0	device_current
Is(M1):	-2.79511e-05	device_current
Is(M2):	-2.78263e-05	device_current
Is(M3):	-2.79511e-05	device_current
Is(M4):	-2.78263e-05	device_current
Is(M5):	-4.68687e-05	device_current
Is(M6):	-4.68686e-05	device_current
Is(M7):	-5.57773e-05	device_current
Is(M8):	-4e-06	device_current


ota2ndstageprojectdipyaman240102116
V(n004)
1.38V
1.32V
1.26V
1.20V
1.14V
1.08V
1.02V
0.96V
0.90V
0.84V
0.78V
0µs
2µs
4µs
6µs
8µs
10µs
12µs
14µs
16µs